University of Colombo, Sri Lanka

UCSC *University of Colombo School of Computing* BIT

DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY (EXTERNAL)

Academic Year 2022 — 2nd Year Examination — Semester 3

IT3206 — Data Structures and Algorithms

Part 1 - Multiple Choice Question Paper

(One hour)

Important Instructions

- This paper has **two (2) parts**, **Part 1** and **Part 2**.
- The duration of this paper (Part 1) is **One (1) hour**.
- The medium of instructions and questions is English.
- This paper has **25 questions** on **9 pages**. Answer **all** questions.
- All the questions in this part 1 are of the **MCQ** (Multiple Choice Questions) type.
- Each question will have **5 (five)** choices with **one or more** correct answers.
- All the questions will carry equal marks.
- There will be a penalty for incorrect responses to discourage guessing.
- The mark given for a question will vary from -1 (All the incorrect choices are marked & no correct choices are marked) to +1 (All the correct choices are marked & no incorrect choices are marked). However, **the minimum mark per question would be zero**.
- Answers should be marked on the **special answer sheet** provided.
- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor/invigilator immediately.
- Mark the correct choices on the question paper first and then transfer them to the given answer sheet which will be machine marked. **Please completely read and follow the instructions given on the other side of the answer sheet before you shade your correct choices.**
- Calculators are **not** allowed.
- *All Rights Reserved.* This question paper can NOT be used without proper permission from the University of Colombo School of Computing.

1) Select an application(s) of the stack data structure.

I Arithmetic expression evaluation
II. Handle function calls
III. To manage memory and storage
IV. For backtracking
V. To implement hashes

(a) I, II, and III Only	(b) I, II, and IV only	(c) I,II,III,and IV only
(d) I, II, IV and V Only	(e) I, II, III, IV and V	

2) Which data structure is identical to *separate chaining* upon collision in a hash node in hash maps?

(a) Queue	(b) Array	(c) Stack
(d) Linked List	(e) Tree	

3) A list has connections to one direction, and the last node of the list connects to the first node of the list. This data structure is called a

(a) Circular Linked List	(b) Singly Linked List	(c) Doubly Linked List
(d) Heap	(e) Array	

4) Which of the following statement(s) is/are correct regarding a Linked List?

I. A linked list provides a linear representation of links from one node to another.
II. A doubly linked list is a circular list that contains previous and next pointer nodes.
III. A circular linked list connects nodes as a circle with no null at the end.

(a) I only	(b) I and II Only	(c) I and III Only
(d) II and III Only	(e) I, II, and III.	

5) Consider the following infix expression
3+5+(4*(6*3)-11)
If one converts the above expression into (i) postfix and (ii) prefix, what would be the correct expressions after conversion?

(a) (i) 354+63**11-+ (ii) +3+5-*4*6311
(b) (i) 35+463**11-+ (ii) +3+5-*4*6311
(c) (i) 35+463**1-1+ (ii) +3+5-*46*311
(d) (i) +3+5-*4*6311 (II) 35+463**11-+
(e) (i) 35+4*63*11-+ (ii) +3+5-*46*311

6) Consider the following stack operations performed on a stack of size 5.

```
Push(1);
print (Pop ());
print (Is_empty())
Push(2);
Push(3);
Print(Pop());
Prnt(Is_full());
Push(4);
Print(Pop());
Print(Pop());
Print(push(5));
Print(is_empty());
```

What would be the results when performing the above series of operations?

(a) 1, true,2, false, 4,2, true
(b) 1, true, 3, false,4,2,5, true
(c) 1, true, 3, false,4,2,5, false.
(d) 1, true,3, false,4,2, false
(e) 1, false,3, false,4,2, false

7) Consider the following pseudo code algorithm segment.

```
// number is an integer value
Create an empty stack
Read (number)
Loop (number > 0)
  digit = number % 2
  push (digit)
  number = number // 2 (integer division)
pop all the digits from the stack
```

What would be the series of popped values if number=21?

(a) 12	(b) 10101	(c) 11101
(d) 21	(e) 11001	

8) Consider the following pseudocode algorithm

```
Step 1: IF REAR =MAXSIZE-1
                WRITE OVERFLOW
                GOTO SETP 4
            [END OF IF]
Step 2: IF FRONT=-1 AND REAR=-1
                SET FRONT=REAR=-0
            ELSE
                SET REAR=REAR+1
            [END OF IF]
Step 3: SET PROCEDURE [REAR]=NUM
Step 4: Exit
```

What is/are the valid statement(s) from the following related to the above pseudocode?

(a) Describes how to insert a new value into a queue.
(b) Describes how to delete the frontmost element from the queue.
(c) Describes inserting a new value into an empty and non-empty queue.
(d) Describes how to delete a value from a circular queue.
(e) The most appropriate boundary-checking condition is included in the pseudocode.

9) What will be the output if the following recursive function was called with input parameter **7** for the **"number"?**

```
public static void MyFunction(int number){
      if (number > 0){
            System.out.print(number%3);
            MyFunction(number-2);
            System.out.print(number%2);
      }
}
```

(a) 13010101 (b) 12011111 (c) 11111101
(d) 11011111 (e) 12011011

10) Which of the following statement(s) is/are true?

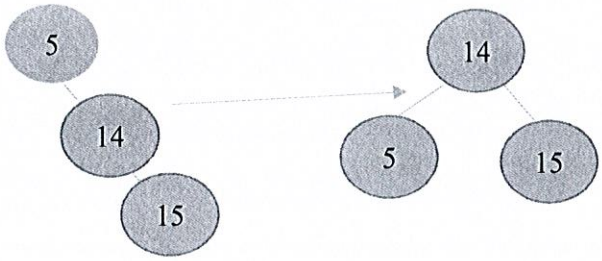
(a) Java uses the Stack data structure to manage recursion.
(b) Better time complexity is one of the main advantages of using recursive solutions.
(c) A recursive solution can always be replaceable with a non-recursive solution.
(d) Recursive solutions have a better space complexity than non-recursive solutions.
(e) Recursive solutions divide a complex problem into sub-problems of the same form.

11) Which of the following statement(s) is/are accurate about tree data structure?

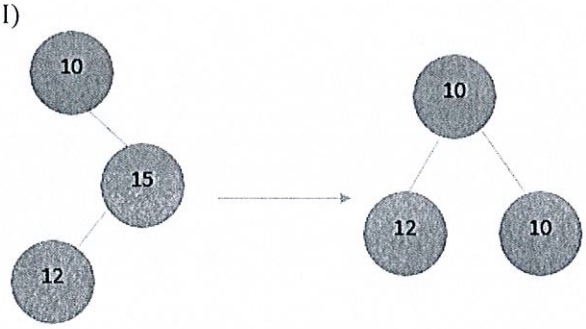
(a) If a tree has N nodes then it will have (N-1) edges
(b) There is only one path from each node to any other node of the tree.
(c) In some occasions, multiple paths exist from one node to another node.
(d) The total count of subtrees attached to that node is called the degree of the node
(e) The general trees can be implemented with a minimum of two pointers.

12) Consider the following.

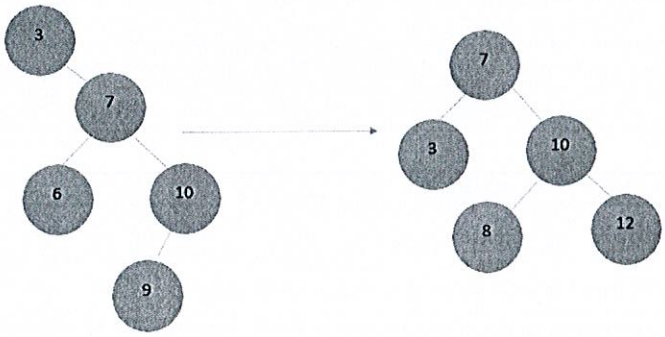
I)



(II)



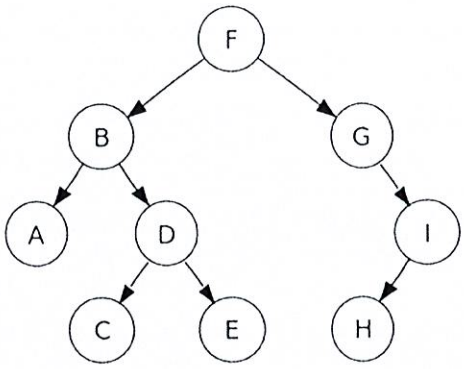
(III)



Which of the above are correct rotations used to establish an AVL-property violation?

(a) (I) Only	(b) (I) and (II) Only	(c) (II) Only
(d) (II) and (III) Only	(e) All (I), (II) and (III)	

13) What is the correct post-order traversal for the following tree?



(a) ABDCEGFIH	(b) ACEDBHIGF	(c) ACDEBIHGF
(d) ABCDEFGHI	(e) CDEBAGIHF	

14) What will be the worst-case time complexity for searching an item in a Binary Search Tree?

(a) $O(n^2)$	(b) O(n)
(c) O(nlogn)	(d) O(logn)
(e) $O(\log n^2)$	

15) Which of the following statements are correct regarding Depth First Search (DFS) algorithm?

I. Explore all vertices and edges of the given graph (G)
II. Search algorithm runs until all nodes are visited.
III. Backtracking is used to discover visited last node to explore more unvisited nodes.
IV. DFS can be used on both directed and undirected graphs.

(a) I and II Only	(b) I and III only	(c) I, II, and III Only
(d) II, III and IV Only	(e) I, II. III and IV	

16) For an undirected graph G with n vertices and e edges, the sum of the degrees of each vertex is:

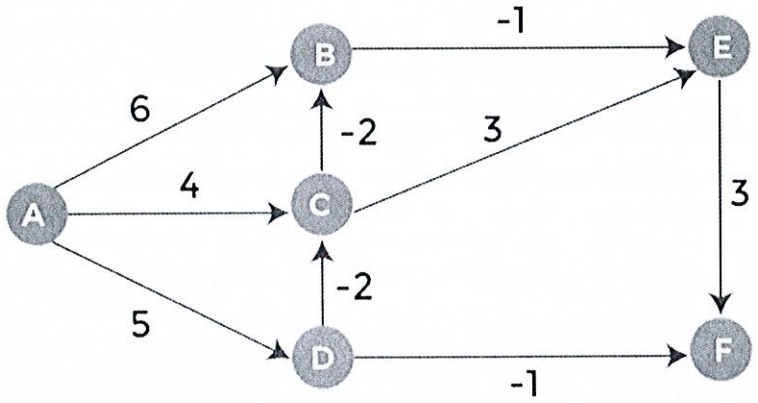
(a) 4e	(b) ne	(c) 2n
(d) e^n	(e) 2e	

17) Which of the following statements are correct regarding Dijkstra's algorithm?

I. Given a weighted digraph and source vertex S, we can find shortest path from S to an arbitrary node t.
II. Works for both directed and undirected graphs.
III. Works with negative edges.
IV. Only works for connected graphs.

(a) I and II Only	(b) I and III only	(c) I, II, and III only
(d) I, II and IV Only	(e) II, III and IV Only	

18) What is/are the shortest path/s from A node to F node?



(a) A→D→F	(b)A→B→E→F	(c)A→C→B→E→F
(d) A→C→ E→F	(e) A→D→C→B→E→F	

19) Which of the following statement/s is/are true?

(a) The running time of Dijkstra's algorithm is O(n Log(n))
(b) Dijkstra's algorithm is used to solve multiple source shortest path problems.
(c) Dijkstra's algorithm cannot be applied to graphs having negative weights
(d) Bellman-Ford algorithm returns an Integer value.
(e) The Bellman-Ford algorithm can find the shortest paths in graphs with negative weight cycles

20) What is the time complexity of the following code segment?

```
for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if(j < 1){
                    System.out.println(j);
                }else {
                 Break;
                }
            }
        }
```

(a) $O(n^2)$	(b) O(n)	(c) O(nlogn)
(d) O(logn)	(e) $O(logn^2)$	

21) What is the best-case time complexity for Linear Search?

(a) O(1)	(b) O(n)	(c) O(nlogn)
(d) O(logn)	(e) $O(logn^2)$	

22) Which of the following statement/s is/are true?

(a) Algorithms A and B have a worst-case running time of O(n) and O(logn), respectively. Therefore, algorithm B always runs faster than algorithm A
(b) Time complexity is measured by counting the number of primitive operations the algorithm performs on a given input size
(c) Space complexity is a more useful criterion for comparing the efficiency of algorithms than Time complexity
(d) Asymptotic analysis is a better measure of the efficiency of an algorithm than Execution time.
(e) The big-O notion is used to describe the average-case time complexity of an algorithm.

23) The Selection Sort algorithm works as follows:

In terms of Array A, the selection sort finds the smallest element in the list and exchanges it with A [0]. Then, ignoring A [0], the sort finds the next smallest and swaps it with A [1] and so on.

Which of the following is **not** an intermediate outcome of the selection sort algorithm that works on the following sample data set?

Sample Data: [15, 47, 39, 27, 59, 03, 65, 21, 70]

(a) 03,15,21,27,59,47,65,39,70
(b) 03,15,21,27,59,47,39,65,70
(c) 03,47,39,27,59,15,65,21,70
(d) 03,15,39,27,59,47,65,21,70
(e) 03,15,21,59,27,47,65,39,70

24) Which of the following is **incorrect** related to the heap, heap construction and the heap sort?

(a) After the heap construction, the sorting is done by using the following techniques. (i) **Swapping** the *root element* with the *last element of the array* and decreasing the length of the heap array by one (ii) **Restoring** heap properties (iii) **Repeating** this process until every element in the array is sorted
(b) On a **max heap**, this process will sort the array in **descending order**
(c) On a **min heap**, this process will sort in **ascending order**
(d) The time complexity of heap sort is the same in the best, worst and average cases, and it is O(nlogn)
(e) Heap sort is typically not stable since the operations on the heap can change the relative order of equal key items.

25) Unordered steps of the radix sort algorithm are given below:

Step P: For the first-place value, call counting sort, jump place value by 10 to move to the next significant digit
Step Q: Print out the updated, sorted array
Step R: Find the largest element in the unsorted input array (Array)
Step S: Continue **Step (.....)** for all place values (finish all d passes)
Step T: Exit
Step U: Create a for expression that loops d times, where d = number of digits in the largest element (or maximum)

Which of the correct order of the steps in the algorithm and suitable step number for the blank position in step S?

(a) Step R, Step U, Step P, Step S, Step T, Step Q and P.
(b) Step R, Step U, Step P, Step S, Step Q, Step T and U.
(c) Step R, Step U, Step P, Step S, Step Q, Step T and P.
(d) Step R, Step U, Step S, Step P, Step Q, Step T and P.
(e) Step R, Step Q, Step P, Step S, Step U, Step T and P.

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